Lesson Plan

Teacher Name: Ms. Lubna Kakkengal and Mr. Dilan Devendra	**Class/Section:** Grade 4
Subject: TCD: Matatalab Pro Coding	**Date: Week 12 (16 Nov to 20 Nov)**
Class Description: Grade 4 TCD class, Matatalab Pro Session 3 of 4 (1 hour 30 minutes). Using the map mat and obstacles, students plan and code a multi-step mission route, test it and troubleshoot any mismatch between their code and the robot's movement.	
TA/LSA Role (if applicable):	

Differentiated Learning Groups (Data)			
Desert Fox	**Camel**	**Oryx**	**Falcon**

Lesson Overview		
Lesson Topic	Mission Route Challenge	
Standard(s)	Technology: plan, test and debug algorithms; Math: 4.MD.C.5 (angles), grid coordinates	
Learning Objective(s) (WALT): We are learning to plan, code and debug a multi-step route around obstacles.		**Success Criteria (WILF):** I can plan a route on a grid before I code it. I can use loops, functions and angle blocks in one program. I can find the exact block where my code went wrong. I can record my test and my fix in a log.
Literacy Strategies	Route planning sheet with grid letters and numbers; debugging log with sentence frames: The robot went … but I wanted … so I changed …	
Assessment	Route plan and debugging log; mission success; live quiz on functions, angles and loops on the TCD website.	
Targeted Vocabulary	route, grid, obstacle, loop, function, test, bug, debug log	
Skills	Planning, spatial reasoning, debugging, perseverance	

National Identity Integration	Plan a route through a map of a UAE city: start at the Dubai Frame, avoid the obstacles and finish at the Museum of the Future.
AI/Digital Learning	Matatalab Pro Sets; hook video; live quiz on the TCD website. Presentation: Grade4_Week12_Presentation.pptx. Worksheet: Grade4_Week12_Worksheet.

Lesson Structure	
Starter (Hook) Engage and Ask (10 minutes)	Warm-up: Grid Treasure (slide 4). Show the hook video and ask the hook question (slide 2). Share the learning goals (slide 3) and key words (slide 5).
I Do (15 minutes)	Model route planning on the grid sheet (slide 6). Recap loops and functions and when to use each (slide 7). Model debugging with a deliberate mistake (slide 8).
We Do (15 minutes)	Class plans one short route together on the board and predicts the code. Pairs set up their mat with landmarks and obstacles (slide 9).
You Do (40 minutes)	Pairs plan, code, test and debug their mission (slide 11). Challenge levels on the board (slide 12). Live quiz on functions, angles and loops (slide 13).
Differentiated Instruction: **Beginning / On / Above**	**Beginning: Route with two obstacles and a planning sheet with the turns already marked.** **On: Route with four obstacles, using a loop or a function.** **Above: Route that visits two landmarks before the finish, using a loop and a function, in the fewest blocks.**
Closure/ Reflection (10 minutes)	Pairs share one bug they found and how they fixed it. Exit ticket (slide 14). Pack away.

Inclusive Planning
Students receiving additional inclusive support (IEP or ILPs) or Gifted and Talented: EAL: debugging log sentence frames and landmark picture cards. SEN: fewer obstacles and pre-marked turns. G&T: two-landmark route in the fewest blocks.
Inclusion / EAL Support

☑ Visual Supports
☑ Chunked Instructions
☐ Reduced Language Load
☐ Model / Worked Example
☐ Pre-teach Vocabulary
☐ TA/LSA Support
☐ Extra Processing Time
☐ Text Read by TA/LSA
☑ Flexible Grouping
☐ Assistive Technology
☐ Movement Break
☑ Scaffold / Sentence Starters
☐ Other:

G&T / Extension

☑ Extension / Challenge Task
☑ Higher-order Questioning
☐ Depth & Complexity Task
☐ Independent Inquiry
☐ Leadership / Peer Coaching
☐ Real-World Application
☐ Curriculum Compacting
☐ Above-grade-level Texts
☐ Student-led / Passion Project
☑ Open-Ended Task
☐ Other:

Assessment Adaptations

☑ Oral Response
☐ Reduced Questions
☑ Scaffolded Success Criteria
☐ Extra Time
☐ Alternative Outcome Format
☑ Teacher Conferencing/Live Feedback
☐ Quiet / Separate Space
☐ Other: